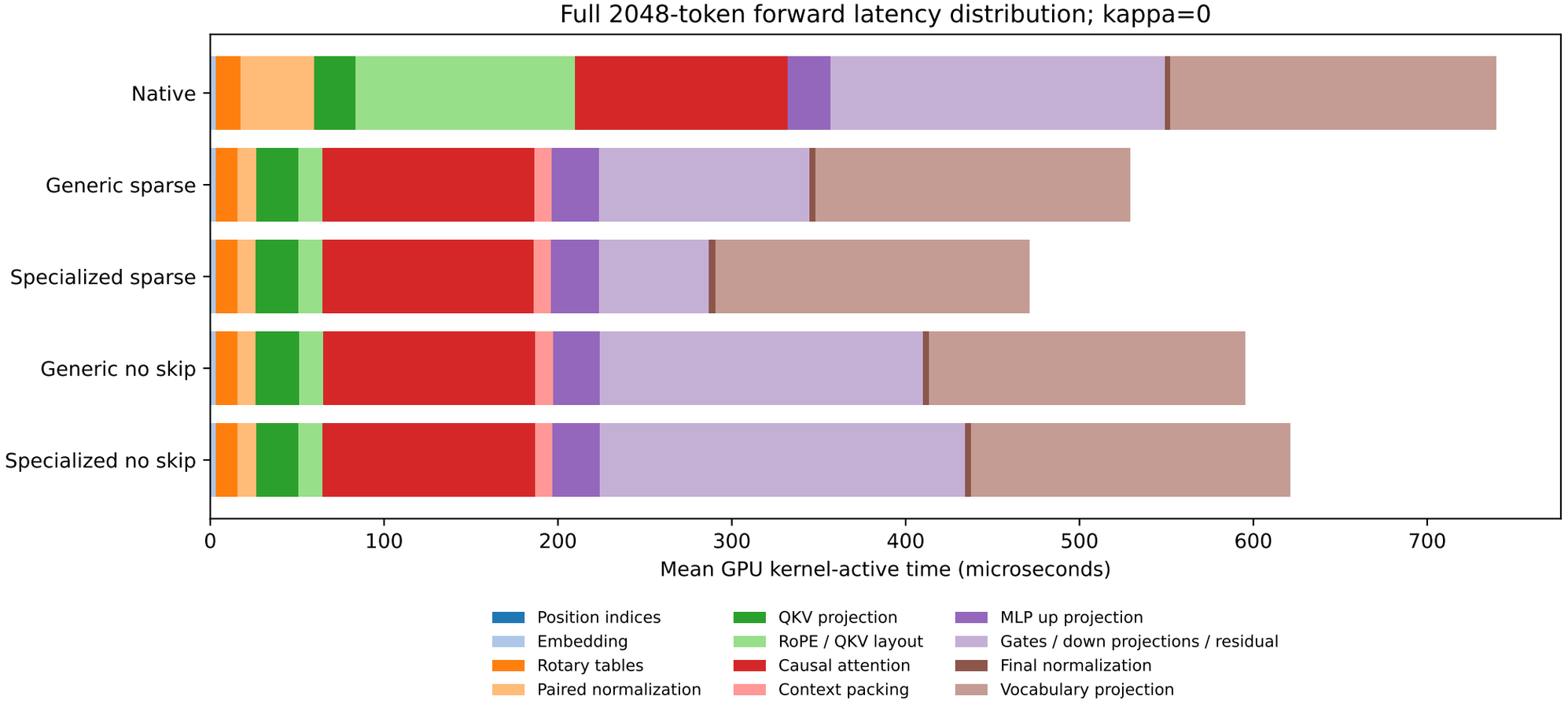
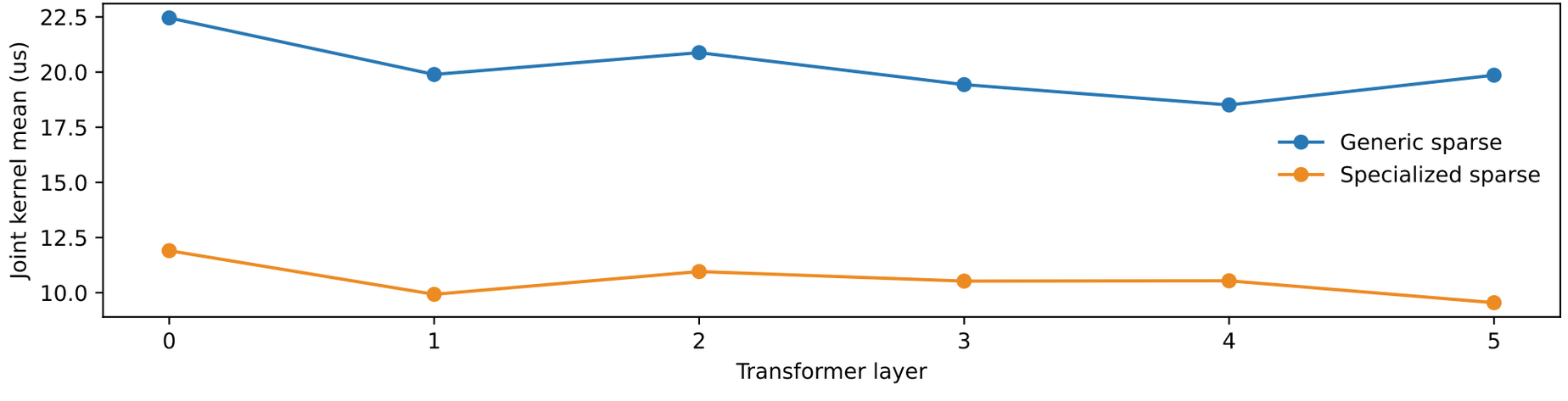
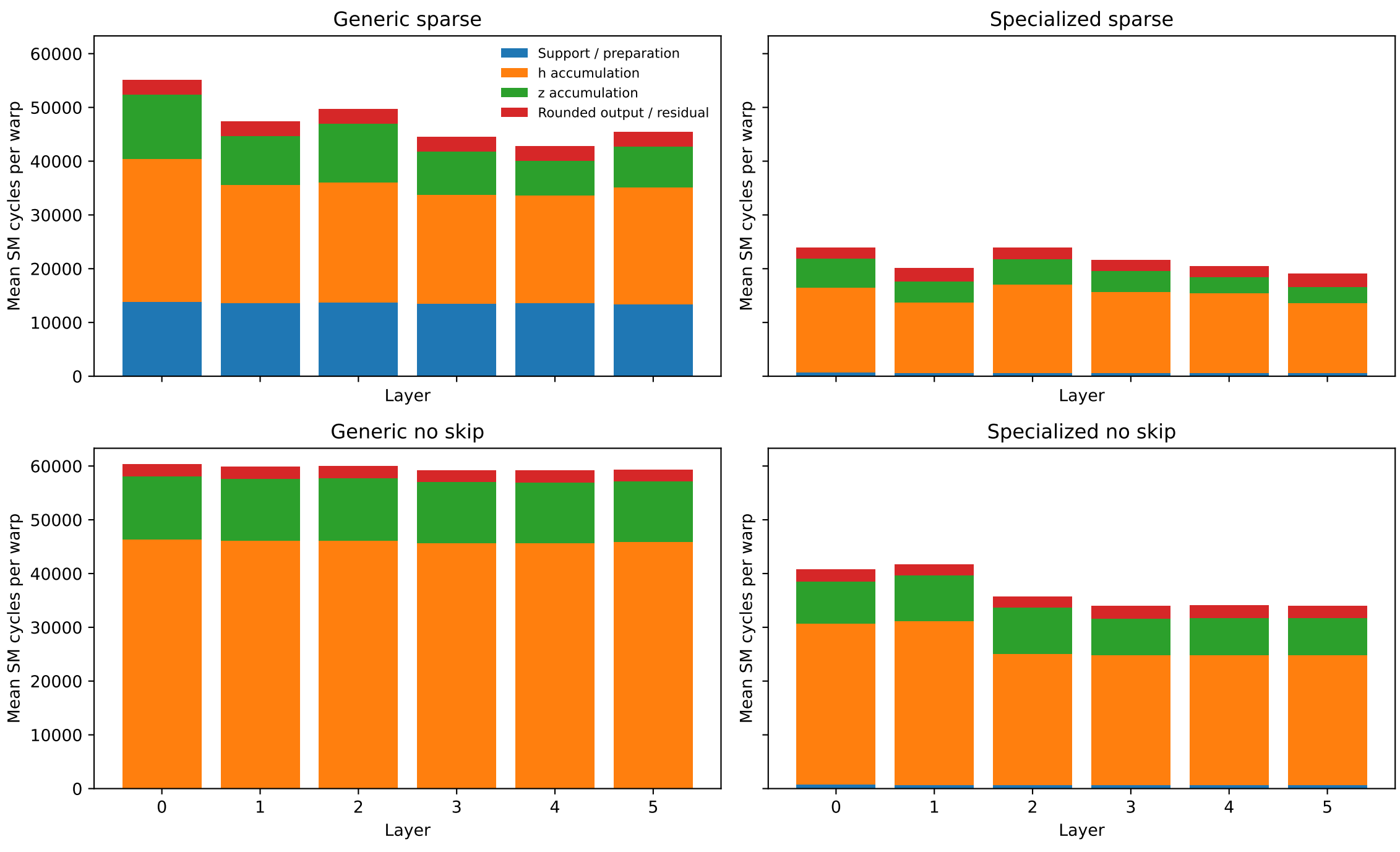
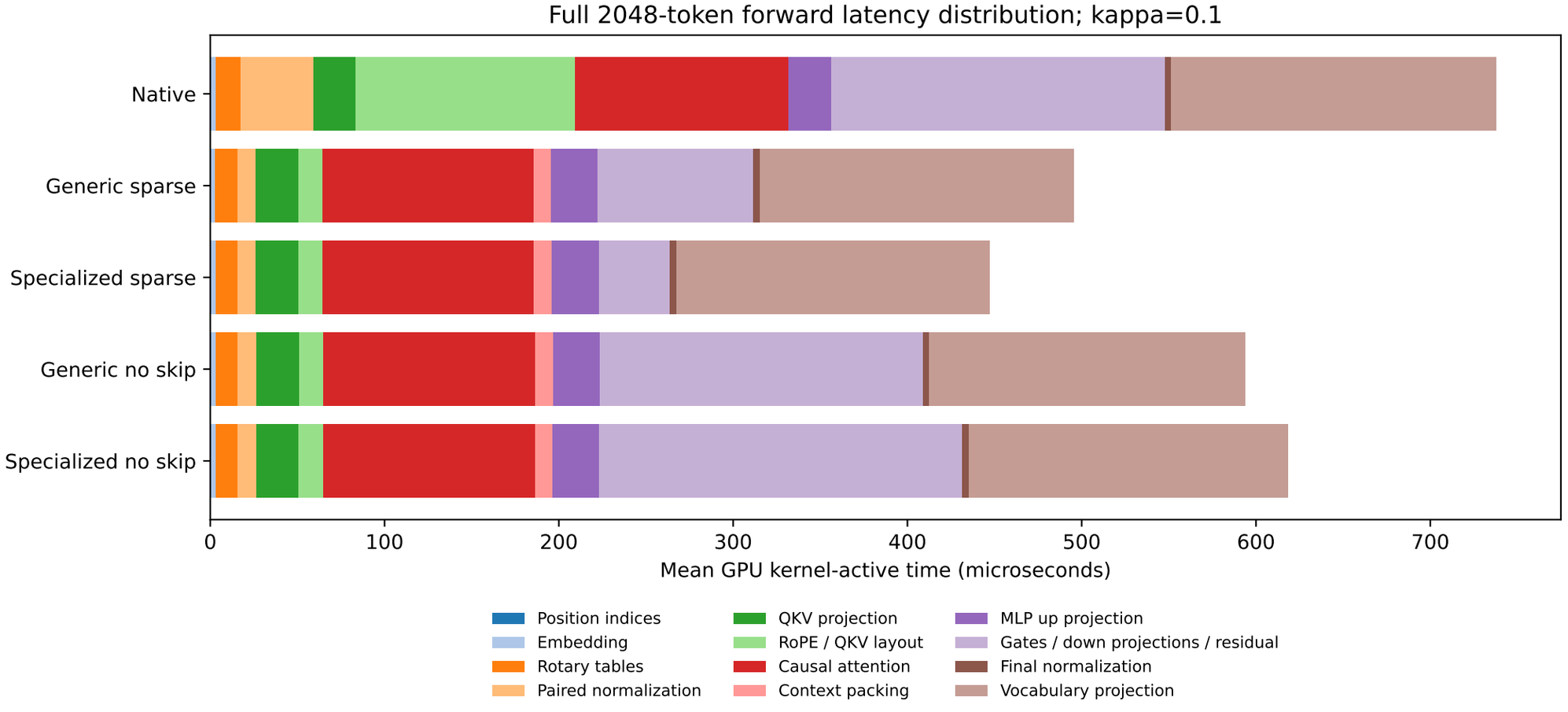
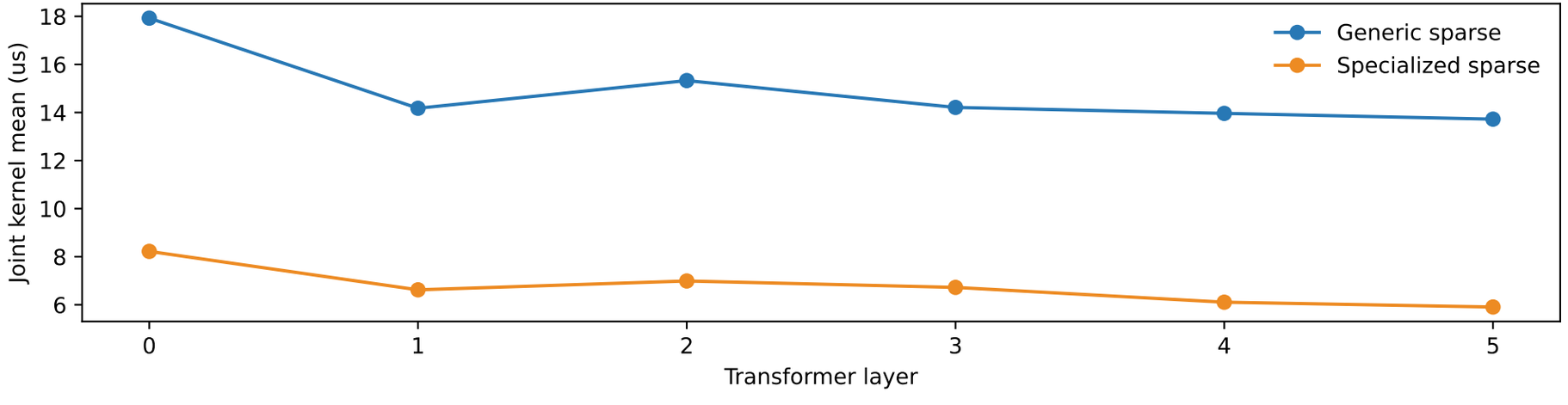
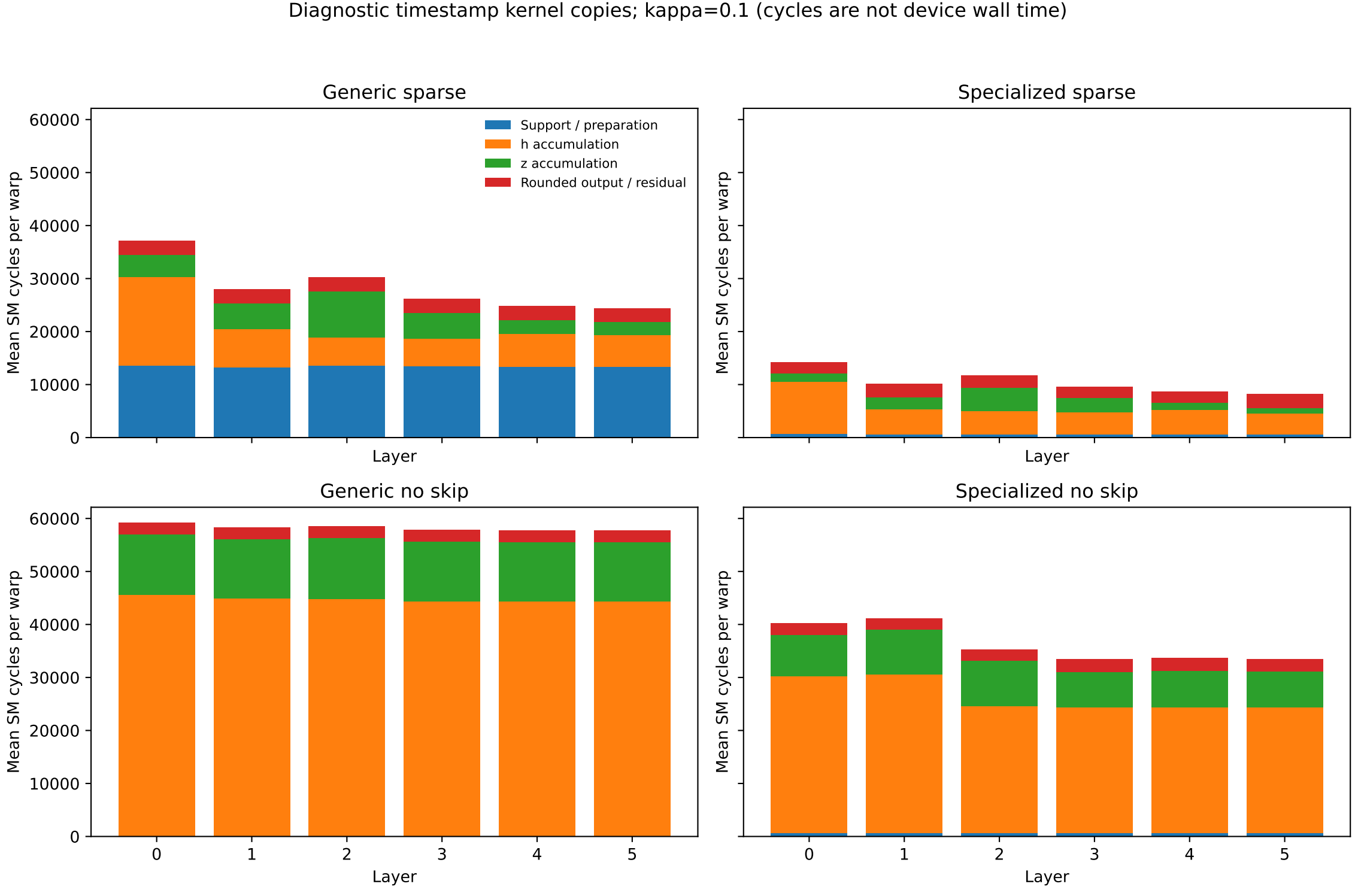

Full 2048-token forward latency distribution; kappa=0
Native
Generic sparse
Specialized sparse
Generic no skip
Specialized no skip
0
100
200
300
400
500
600
700
Mean GPU kernel-active time (microseconds)
Position indices
Embedding
Rotary tables
Paired normalization
QKV projection
RoPE / QKV layout
Causal attention
Context packing
MLP up projection
Gates / down projections / residual
Final normalization
Vocabulary projection


Joint kernel mean (us)
22.5
20.0
17.5
15.0
12.5
10.0
0
1
2
3
4
5
Transformer layer
Generic sparse
Specialized sparse


Diagnostic timestamp kernel copies; kappa=0 (cycles are not device wall time)
Generic sparse
Specialized sparse
Generic no skip
Specialized no skip
Support / preparation
h accumulation
z accumulation
Rounded output / residual
Mean SM cycles per warp
Layer
0
10000
20000
30000
40000
50000
60000
0
1
2
3
4
5


Full 2048-token forward latency distribution; kappa=0.1
Native
Generic sparse
Specialized sparse
Generic no skip
Specialized no skip
0
100
200
300
400
500
600
700
Mean GPU kernel-active time (microseconds)
Position indices
Embedding
Rotary tables
Paired normalization
QKV projection
RoPE / QKV layout
Causal attention
Context packing
MLP up projection
Gates / down projections / residual
Final normalization
Vocabulary projection


Joint kernel mean (us)
18
16
14
12
10
8
6
0
1
2
3
4
5
Transformer layer
Generic sparse
Specialized sparse


Diagnostic timestamp kernel copies; kappa=0.1 (cycles are not device wall time)
Generic sparse
Specialized sparse
Generic no skip
Specialized no skip
Support / preparation
h accumulation
z accumulation
Rounded output / residual
Mean SM cycles per warp
Layer
0
10000
20000
30000
40000
50000
60000
0
1
2
3
4
5